

# AE-331: NMR of [Ru(CO)(OH)2(PNPtBu)] (AE-310-2) at lower and higher Temperature

Date: 2024-07-25

Tags: [Ru(CO)(OH)2(PNP)] NMR AE  
reference analytics 1H 31P 1H-31P-  
HMBC deep temperature NMR 1H-1H-  
NOESY 1H-1H-ROESY 1H-DOSY

Status: Done

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## Reaction scheme/sample structure

## Reagents

| Name                  | CAS Number / Experiment Number                                                              | Amount [mmol] | Equivalents | Mass <sub>theo</sub> [mg] | Mass <sub>exp</sub> [mg] | Molar mass [g/mol] | Volume [ml] |
|-----------------------|---------------------------------------------------------------------------------------------|---------------|-------------|---------------------------|--------------------------|--------------------|-------------|
| [Ru(CO)(OH)2(PNPtBu)] | Experiment - AE-310:<br>Synthesis of<br>[Ru(CO)(OH)2(PNPtBu)] using<br>Ag2O, 5 h (AE-310-2) | /             | /           | 10                        | 9.82                     |                    | /           |
| MeCN-d3               | 2206-26-0                                                                                   | /             | /           | /                         | /                        | /                  | 0.95        |

## Procedure/observations

All steps (unless stated otherwise) were carried out under argon using standard [Protocol - Schlenk Technique](#)

Since the compound is light sensitive the steps were carried out with the light turned out in the fumehood and the lab.

| Date  | Time  | Step                                                                                                                                                                                        | Observations           |
|-------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 25.07 |       | A 10 mL Schlenk flask was prepared                                                                                                                                                          |                        |
|       | 11:05 | To the flask [Ru] was added in air                                                                                                                                                          |                        |
|       |       | The flask was set under argon                                                                                                                                                               |                        |
|       | 12:55 | To the flask MeCN-d3 (0.95 mL) was added                                                                                                                                                    | orange, turbid mixture |
|       |       | The mixture was stirred                                                                                                                                                                     |                        |
|       | 12:58 | The mixture was filtered using a PTFE syringe filter (pore diameter = 0.2 µm)                                                                                                               |                        |
|       |       | The filtrate from the syringe was transferred into the transition piece of a Young type NMR tube and the NMR was prepared according to <a href="#">Protocol - Preparation of NMR Sample</a> | clear orange solution  |

## Analysis/Results

Measured on 500 MHz IOMC:

[2024-08-08\\_9-22-09.zip\\_AE-331-1\\_10.nmrium](#)

<sup>31</sup>P and <sup>31</sup>P/<sup>1</sup>HMBC: measured at rt, as expected no interesting features observed

<sup>1</sup>H: measured at rt, 0, -10 and 50 °C: H<sub>2</sub>O peak shifts, but normal according to NMR department

NOESY: measured at rt, 0, -10 and 50 °C: interesting cross signal at 5.5 and 6.1 ppm --> could be from OH protons, but no other coupling of this protons observed

DOSY: at rt: signal for complex at -8.95 to -8.92, equals diff. coefficient (D).

Used for dyn. visco. (eta):

<https://www.carlroth.com/medias/SDB-2216-DE-DE.pdf?context=bWFzdGVyfHNIY3VyaXR5RGF0YXNoZWV0c3wzMDcxOTh8YXBwbGljYXRpb24vcGRmfGg2Ny9oMDUvOTE0NDAwMDYxMDMzNC9TREJfMjlxNI9ERV9ERS5wZGZ8Yjc0OWJiY2IxODFjMTQ1NTI5YWMxOWNmYjRIMDEzNGRiZTQyOWE4ODA1NTFhMDUyNDhhYjM0OTkxYmRiYzc4Yw> (SDB from MeCN) deuterated was not found.

T from NMR Data

calculation done:  $r_H = k_B \cdot T / (6 \pi \cdot \eta \cdot D)$

Result: 5.351 +- 0.19 Å

Duterated found:

<https://www.carlroth.com/medias/SDB-7911-GB-EN.pdf?context=bWFzdGVyfHNIY3VyaXR5RGF0YXNoZWV0c3wyNzk3NTB8YXBwbGljYXRpb24vcGRmfGFEdGhMMmd6T0M4NU1UYzVOREkyTIRnMk5qVTBMMU5FUWw4M09URXhYMGRDWDBWT0xuQmtaZ3xmNGVkdGFjMGQxYTQwZWVhMDUyNDhhYjM0OTkxYmRiYzc4Yw> &utm\_source=chatgpt.com

is 0.39 mPa \* s --> hydrodyn-radius = approx. 4.8 Å +- 0.17

## Next Steps:

Sample in MeCN-d<sub>3</sub> with no residual water, retest NOESY, DOSY

## Linked experiments

- [AEI-078: Degassing of DCM-d2](#)
- [AE-310: Synthesis of \[Ru\(CO\)\(OH\)2\(PNPtBu\)\] using Ag2O, 5 h](#)
- [AE-317: Synthesis of PNP\(tBu2,S\)\(Bpin\) from PNP\(tBu2,S\)](#)

## Linked resource

Protocol - [Preparation of NMR Sample](#)

## Attached files

AE-321-DOSY.xlsx  
sha256: 358958ad5cd567aa5db3f93735b68f2a1c7869f5e7173e36eec8e67d6b3be706

2024-08-08\_9-22-09.zip\_AE-331-1\_10.nmrium  
sha256: 0107107722a37479f7faa7e8b4e971766487ba001e1dbe5225fde927e9e49d5e

2024-08-08\_9-22-09.zip

2024-08-02\_9-44-20.zip  
sha256: 78d5036a7f8b540c34ee8c5ff94ee352fa29460559cb67f2642b2ade0ce366cc

in-MeCN-d3.jpg  
sha256: 0ddd70cd571e80aea04e316b861124bc61cee57948a558e5a356c8e14b9a622d



NMR-sample.jpg  
sha256: 3e47ced7a9c8bcf1892985617ec42268afe6b8fc491ac4f5ba36c6289aa17beb



## Comment

On 2025-01-10 14:42:45 Jacob Schneidewind wrote:

Notes:

\* Formatting analysis section correctly



Unique eLabID: 20240725-e7f22bf8ed3171a53527c78d14aa38495e9c6e13

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1238>